

Document Title: Solution Requirements	Date: 15 March 2026
Project Name: ComicCraft - AI Comic Story Creator using Gemini Models	Team ID: SWTID-2026-5650
Maximum Marks: 4 Marks	Status: Approved

Define Functional and Non-Functional Requirements

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Priority (High/Med/Low)	Requirement Description
1	Authentication	High	Provide secure user registration, multi-factor login, and social OAuth integration (Google/GitHub) for comic creators.
2	Authorization levels	High	Define role-based access control (Free Creator: 3 stories/mo; Pro Creator: unlimited AI generation & HD PDF export; Admin).
3	External interfaces	High	Integrate with Google Gemini API (1.5 Pro & Flash) for narrative scripting and Gemini/Imagen 3 for image rendering.
4	Transactions processing	High	Manage user prompt submission queues, AI credit consumption, project draft state saving, and canvas layout updates.
5	Reporting	Medium	Deliver analytics dashboards displaying total comic issues created, panel generation latency, storage usage, and community views.
6	Business rules, etc.	High	Enforce AI safety content filtering, maximum 12 panels per story issue, character seed retention for visual consistency, and rate limits.
7	Compliance to laws or regulations	High	Ensure strict GDPR compliance for user personal data, COPPA standards for underage creators, and clear AI content attribution.
8	Other	Medium	Provide drag-and-drop panel reordering, customizable speech bubbles (shapes, fonts, tails), and multi-format export (PDF, CBZ, PNG).

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Optimize response time for AI script creation and visual panel rendering latency.	Script response < 3 sec; image panel render < 5 sec per frame.
2	Scalability	Handle high concurrent user loads and batch image rendering requests during viral traffic spikes.	Auto-scaling microservices supporting 10,000 active concurrent creators.
3	Security & Data Privacy	Detail data protection, API key secret management, encryption, and vulnerability defense.	End-to-end AES-256 encryption at rest, TLS 1.3 in transit, OAuth 2.0.
4	Reliability & Availability	Specify system uptime requirements, fallback mechanisms, and API fault tolerance.	99.9% monthly uptime SLA; exponential retry on Gemini API timeouts.
5	Usability & Accessibility	Provide intuitive drag-and-drop comic editor interface and accessibility compliance.	WCAG 2.1 AA standards compliant; mobile responsive; onboarding < 3 min.
6	Other (Maintainability)	Maintain modular microservice architecture and comprehensive automated test suite.	85%+ unit test coverage, containerized Docker deployment, CI/CD pipeline.